Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/28/2025
PI Name:	Story, Michael D Ph.D.		
Project Title:	Determining Gender Differences in the Incidence of Lung Adenocarcinoma After Space Radiation Exposure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75235-7320	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	09/01/2018	**End Date:**	08/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2024 per JSC Grants Office and NSSC information (Ed., 4/11/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lianghao, Ding Ph.D. (University of Texas Southwestern Medical Center, Dallas)		
Grant/Contract No.:	80NSSC18K1676		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Uncertainties in radiation induced lung cancer risk estimation and its associated mortality rates are among the primary factors limiting the number of safe days an astronaut can spend in space. Initial lung cancer risks are based off epidemiological-based modeling and include cohorts such as the atomic bomb survivor life span study (LSS) whose estimates contain large confidence intervals and whose populations may not reflect astronauts on deep space missions. In order to calculate the permissible exposure limit (PEL) for astronauts it is necessary to collect further information on the risk of lung carcinogenesis due to radiation quality differences (relative biological effectiveness--RBE), sex disparities, and how effective biological countermeasures may reduce or mitigate these risks. The goal of this project is to provide sufficient data to bolster risk estimates and RBE values for lung carcinogenesis from the individual small, intermediate, and heavy charged particles that comprise galactic cosmic rays (GCRs) with doses comparable to what an astronaut may receive on a Mars mission. Additionally we will delineate any sex differences in radiogenic lung cancer risk resulting from space radiation exposure, provide sufficient evidence to validate GC4419, a Food & Drug Administration (FDA investigational new drug (IND), as an effective pharmaceutical countermeasure, and mechanistically define the biological processes associated with space radiation induced lung carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are two areas where this research may benefit life on Earth. 1) Differences in both the incidence of and mortality arising from lung cancer between men and women have long been appreciated, with women generally having higher incidences and mortality rates than men. Epidemiological studies comparing pre- and post-menopausal women treated with hormone replacement therapy (HRT) have demonstrated that female sex hormones both elevated the incidence and aggressiveness at the time of presentation. These effects likely enhance the efficacy of other carcinogens such as radiation or tobacco smoke. 2) GC4419 is a radioprotector for radiation-induced mucositis and lung fibrosis from radiation exposure. What is not known, although the preliminary evidence suggests it could, is whether GC4419 has anti-carcinogenic effects. If it does, the potential to reduce the risk for cancer in humans after environmental or diagnostic radiation exposures is compelling.
Task Progress:	Compiled from final report submitted by the Principal Investigator (Ed., 7/28/25). The data generated from this research project suggests that there is no difference in risk for lung cancer based upon the sex of the animals used in this study. This falls in line with the recent epidemiologic study of radiation workers in the US as part of the million-person study and suggests that estrogen plays no role in lung cancer risk – at least in this animal model. Pathologic characterization of lung tumors does not appear to be differentiated by sex either. This result suggests that genomic analysis of tumors based upon the sex of the host animal would not yield any new information either. However, analysis of circulating miRNA irrespective of sex might have yielded information on the onset of lung cancer. Those serum samples are available for analysis. Probably of greater value overall was the pilot study of GC4419 as a radiation countermeasure. [Ed. Note: See “Gene Expression Profile Analysis As A Prognostic Indicator Of Normal Tissue Response To Simulated Space Radiations”, PI: Story.] Here, with one exception (0.4 Gy of 28Si ions), the use of GC4419 extended the lifetime of the irradiated cohorts as well as the unirradiated animals. One key question remains and that is the potential for increased risk based upon dose rate. Dose rate effects remains a key challenge to determining carcinogenic risk for astronauts. Lastly, GC4419 is given to cancer patients by infusion. This is not an effective approach for astronauts. However, there are now 2 injectable derivatives that should be examined going forward.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Articles in Peer-reviewed Journals	Hosseini S, Sivertz M, Alves EM, Carter LM, Story MD. "Organ dose conversion factors for murine galactic cosmic ray irradiation." Life Sci Space Res. 2026 Feb 5. Online ahead of print. https://doi.org/10.1016/j.lssr.2026.02.001 , Feb-2026